

Problem - 2

$$X = [6.1, 2.8, 4.0, 1.3]$$

$$w_1 = 1, w_2 = 2, w_3 = 4, w_4 = 6$$

$$b = -0.5$$

Formula

$$Z = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b$$

Sol :

Hidden layer

$$Z = (6.1 \times 1) + (2.8 \times 2) + (4.0 \times 4) + (1.3 \times 6) - 0.5$$

$$= 6.1 + 5.6 + 16 + 7.8 - 0.5$$

$$= 35.5 - 0.5$$

$$\underline{\underline{Z = 35.0}}$$

Sigmoid

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-35}}$$

$$\sigma(35) \approx 1 \quad \boxed{h=1}$$

Setosa

$$Z_{\text{setosa}} = 1(1.1) + 0.2$$

$$= 1.3$$

$$\sigma(1.3) = \boxed{0.786}$$

versicolor

$$Z_{\text{versicolor}} = 1(1.4) + 0.1$$

$$Z_{\text{versicolor}} = 1.5$$

$$\sigma(1.5) = \underline{\underline{0.818}}$$

virginica

$$Z_{\text{virginica}} = 1(1.7) + 0.3$$

$$Z_{\text{virginica}} = 2.0$$

$$\sigma(2.0) = \underline{\underline{0.881}}$$